GWODW@RESCEU, UT

Lecture 3:

Parameter Estimation

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Lecture 3: Parameter Estimation
– Content

- Gravitational waveform models
- Bayesian Inference

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- Gravitational waveform models
- Bayesian Inference

Gravitational Waveform Models
– Modeling the compact binary coalescences

- For accurate parameter inference, accurate waveform models must be constructed in advance.

Binary black holes

©Caltech

Binary neutron stars

©KEK

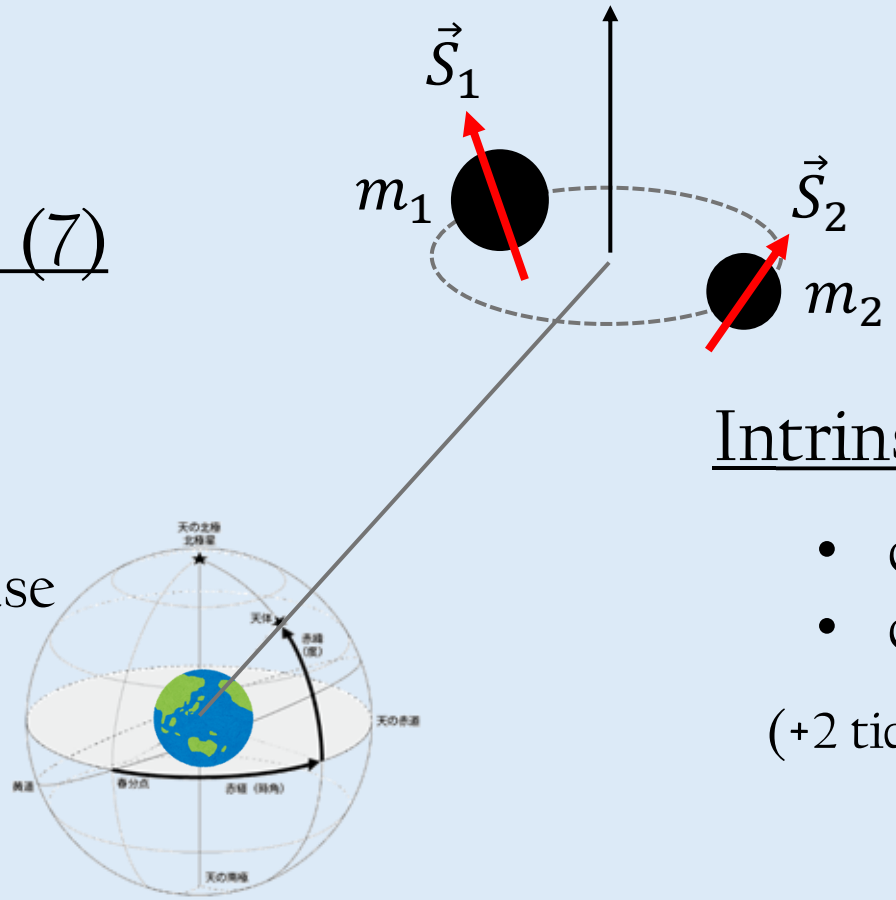
Black hole–neutron star

©C.Knox

Gravitational Waveform Models

– How many parameters are needed for the GW signal

- GW signal is characterized by 15 parameters for binary black holes and 17 for binary neutron stars.



Extrinsic parameters (7)

- luminosity distance
- inclination angle
- sky location (2)
- coalescence time, phase
- polarization angle

Intrinsic parameters (8 or 10)

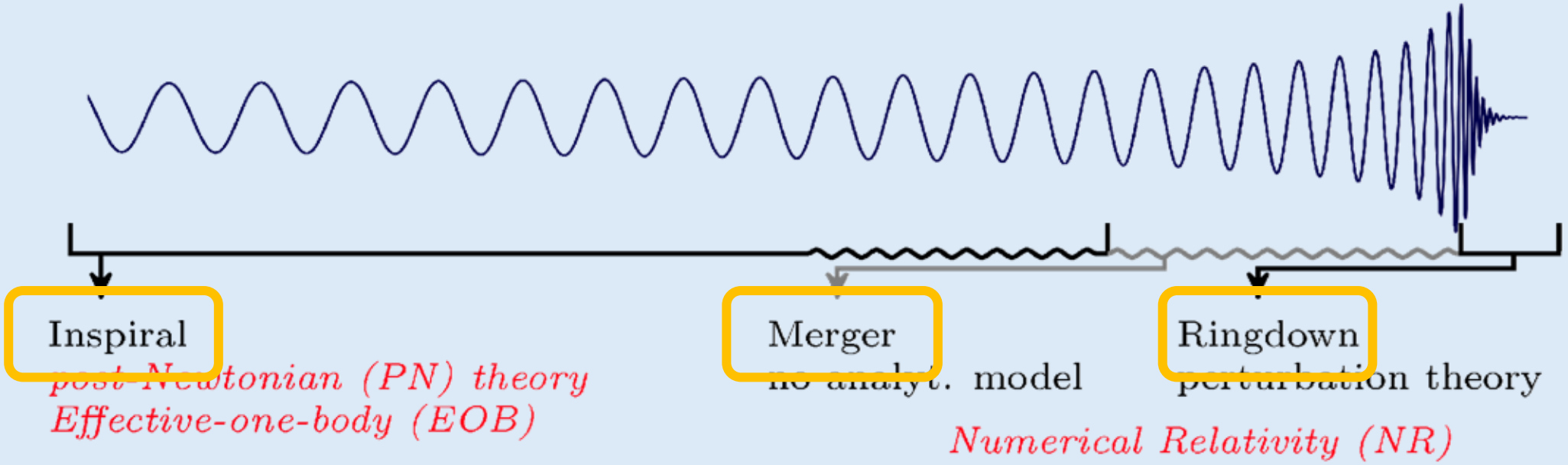
- component masses (2)
- component spins (6)

(+2 tidal deformability in BNS case)

Gravitational Waveform Models

– Coalescence process of a binary

- The coalescence process is commonly divided into 3 stages: the inspiral, merger, and ringdown.



Ohme, 2012

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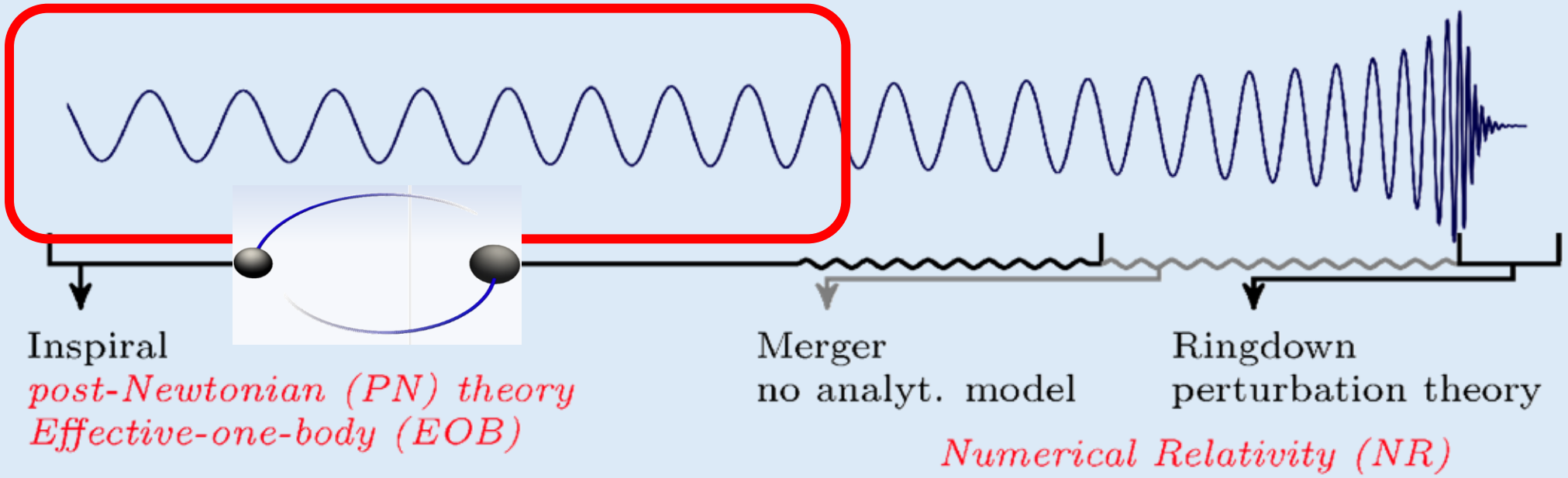

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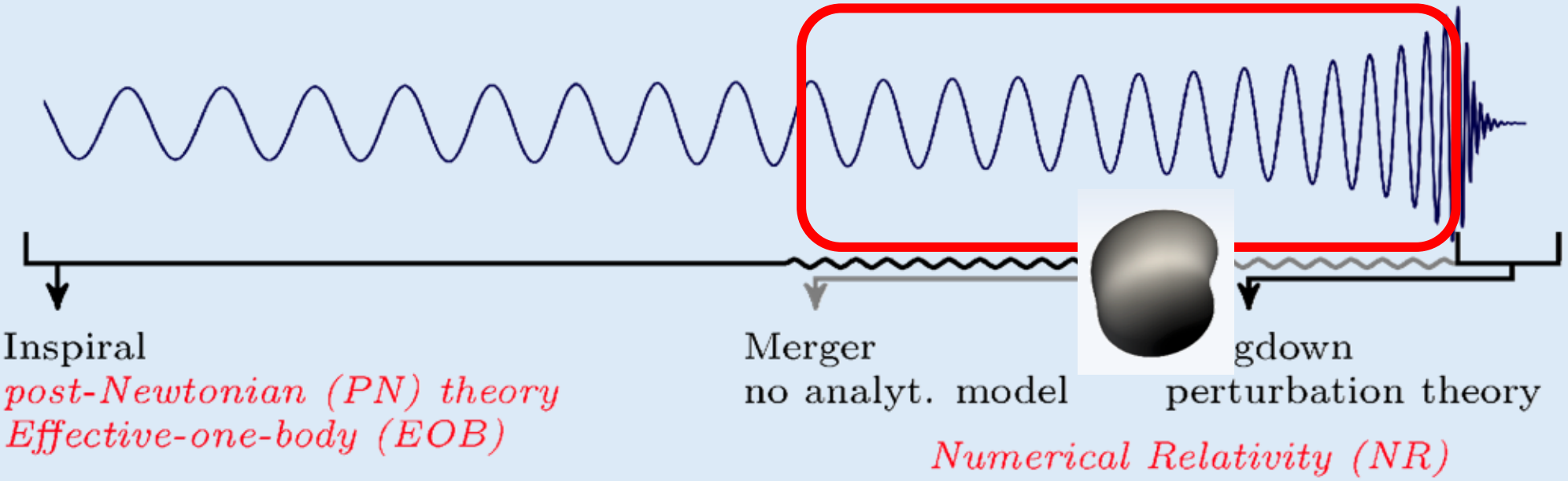

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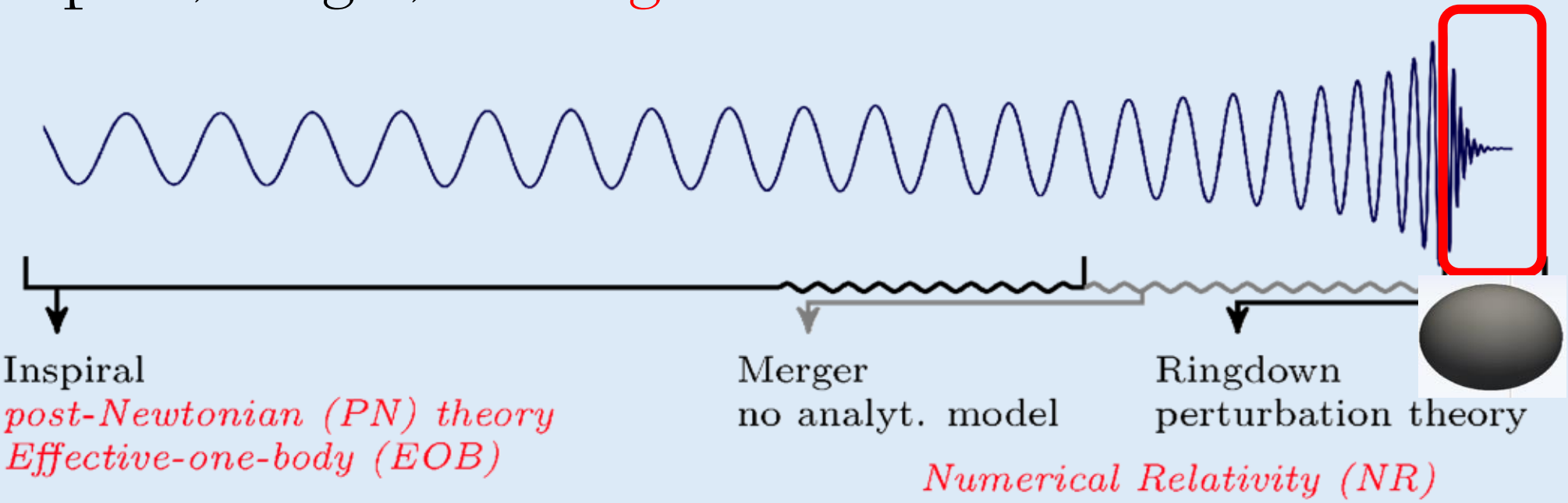

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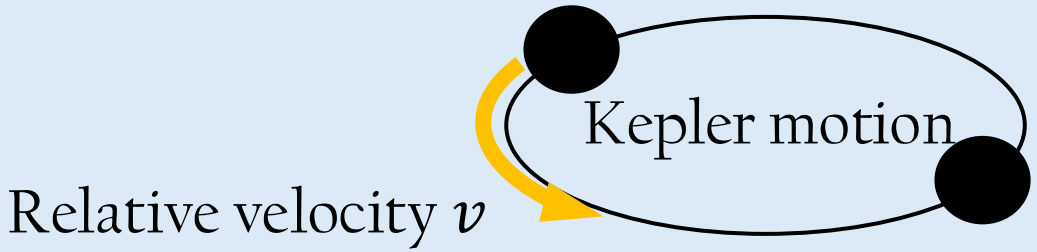
Gravitational Waveform Models

– Modeling the inspiral stage

- The inspiral ≃ Kepler motion + general relativistic correction

<u>Post–Newtonian (PN) expansion</u>



+ GR corrections expressed as an expansion in $\frac{v}{c}$; the exponent n defines the PN order.

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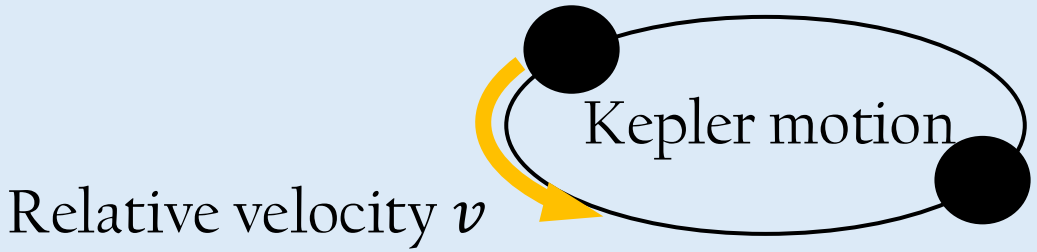

+ GR corrections expressed as an expansion in $\frac{v}{c}$; the exponent n defines the PN order.

- Corresponds to expansion in $f^{\frac{1}{3}}$, where f is GW frequency.

<u>Kepler's 3rd law</u>

$$\frac{r^3}{T^2} = const.$$

$$\left[\begin{array}{c} T = \frac{1}{f} \\ \mathrm{r} = \frac{\mathrm{vT}}{2\pi} = \frac{\mathrm{v}}{2\pi f} \end{array}\right] \quad \rightarrow \quad v \propto f^{\frac{1}{3}}$$

Gravitational Waveform Models

– Modeling the inspiral stage

- The inspiral GW is well modeled by the Post–Newtonian waveform.

$$\tilde{h}_{\mathrm{PN}}(f) = A_{\mathrm{PN}}(f)\mathrm{e}^{\mathrm{i}\phi_{\mathrm{PN}}(f)}$$

$$\boxed{\begin{aligned}\varphi_{\mathrm{PN}}(f) &= 2\pi f t_c - \varphi_c - \frac{\pi}{4} \\ &\quad + \frac{3}{128\eta}\left(\pi\tilde{f}\right)^{-5/3}\sum_{i=0}^{7}\left[\varphi_i + \varphi_{il}\log(\pi\tilde{f})\right]\left(\pi\tilde{f}\right)^{i/3}\end{aligned}}$$

$$\frac{v}{c} = \left(\frac{\pi G M f}{c^3}\right)^{1/3} = \left(\pi\tilde{f}\right)^{1/3}$$: Kepler law

- ϕ_i or ϕ_{il} are the PN coefficients at $i/2$ order.
- Higher PN terms correspond to stronger–gravity effects in general.

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Gravitational Waveform Models

– Modeling the inspiral stage

- The PN coefficients depend on the component masses and spins.

The expansion coefficients are then given by

$$\varphi_0 = 1, \tag{B6}$$

$$\varphi_1 = 0, \tag{B7}$$

$$\varphi_2 = \frac{3715}{756} + \frac{55\eta}{9},$$

$$\varphi_3 = -16\pi + \frac{113\delta\chi_a}{3} + \left(\frac{113}{3} - \frac{76\eta}{3}\right)\chi_s,$$

$$\varphi_4 = \frac{15293365}{508032} + \frac{27145\eta}{504} + \frac{3085\eta^2}{72} + \left(-\frac{405}{8} + 200\eta\right)\chi_a^2 - \frac{405}{4}\delta\chi_a\chi_s + \left(-\frac{405}{8} + \frac{5\eta}{2}\right)\chi_s^2,$$

$$\varphi_5 = [1 + \log(\pi M f)]\left[\frac{38645\pi}{756} - \frac{65\pi\eta}{9} + \delta\left(-\frac{732985}{2268} - \frac{140\eta}{9}\right)\chi_a + \left(-\frac{732985}{2268} + \frac{24260\eta}{81} + \frac{340\eta^2}{9}\right)\chi_s\right],$$

$$\varphi_6 = \frac{11583231236531}{4694215680} - \frac{6848\gamma_E}{21} - \frac{640\pi^2}{3} + \left(-\frac{15737765635}{3048192} + \frac{2255\pi^2}{12}\right)\eta + \frac{76055\eta^2}{1728} - \frac{127825\eta^3}{1296}$$
$$- \frac{6848}{63}\log(64\pi M f) + \frac{2270}{3}\pi\delta\chi_a + \left(\frac{2270\pi}{3} - 520\pi\eta\right)\chi_s,$$

$$\varphi_7 = \frac{77096675\pi}{254016} + \frac{378515\pi\eta}{1512} - \frac{74045\pi\eta^2}{756} + \delta\left(-\frac{25150083775}{3048192} + \frac{26804935\eta}{6048} - \frac{1985\eta^2}{48}\right)\chi_a$$
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S. Khan+, 2016

Parameters of a binary

$$M = m_1 + m_2,$$

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leading spin effect at 1.5 PN

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degeneracies of spin parameters break at 2 PN order

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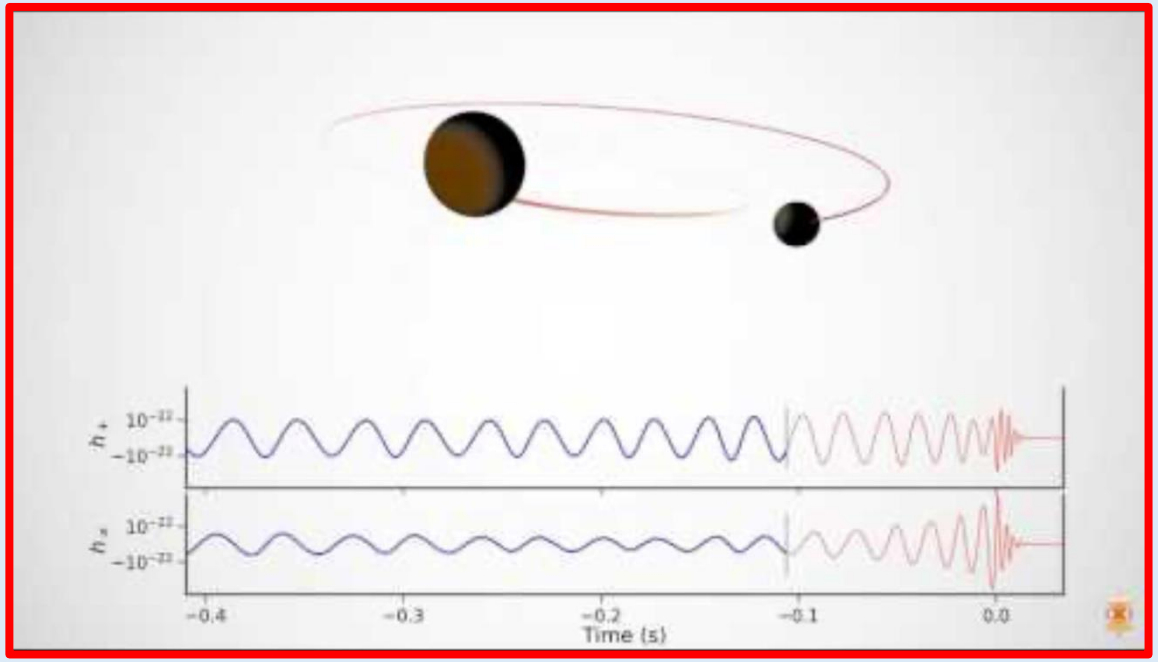
Gravitational Waveform Models

– Modeling the merger stage

- The unique theoretical tool to understand the merger stage is Numerical Relativity.
  - ➢ Solve the Einstein equation numerically and track the nonlinear dynamics.
  - ➢ Difficult to perform for all configurations over parameter space of interest.



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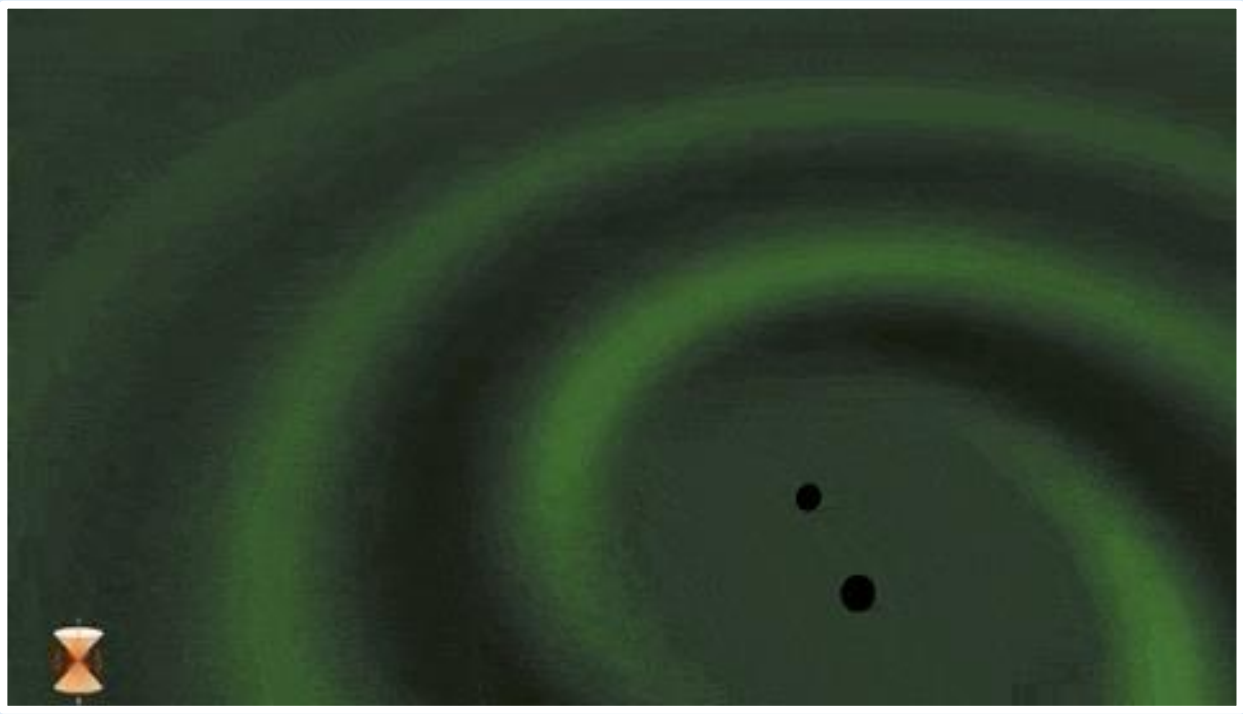
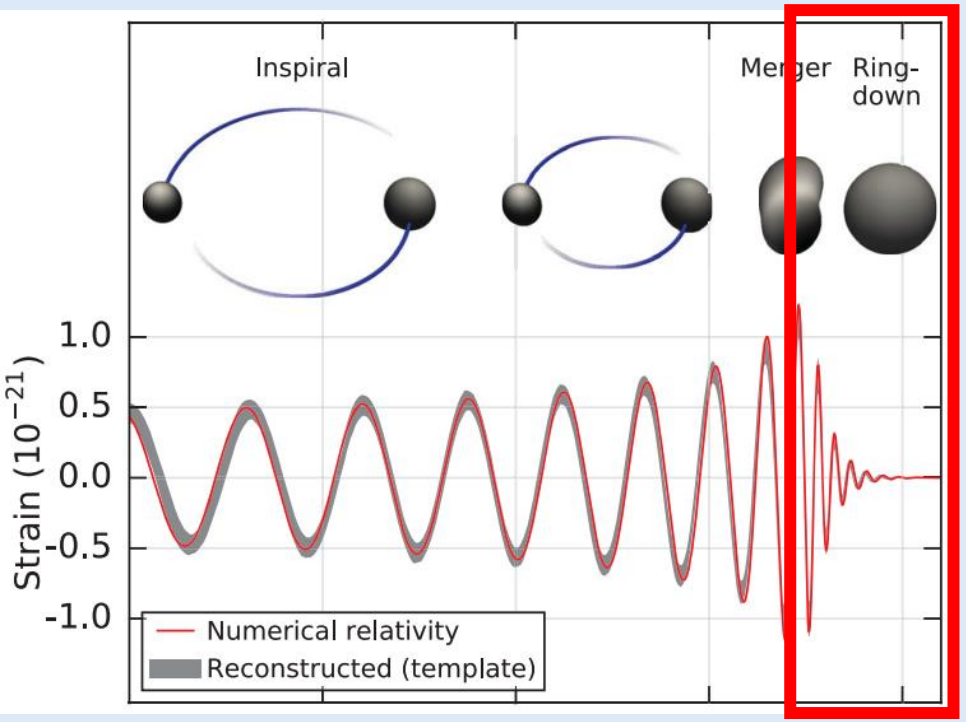
Gravitational Waveform Models

– Modeling the ringdown stage

- Ringdown is the final stage of the coalescence, where the collided BH settles down to a stationary BH (Kerr BH) emitting damping GW.





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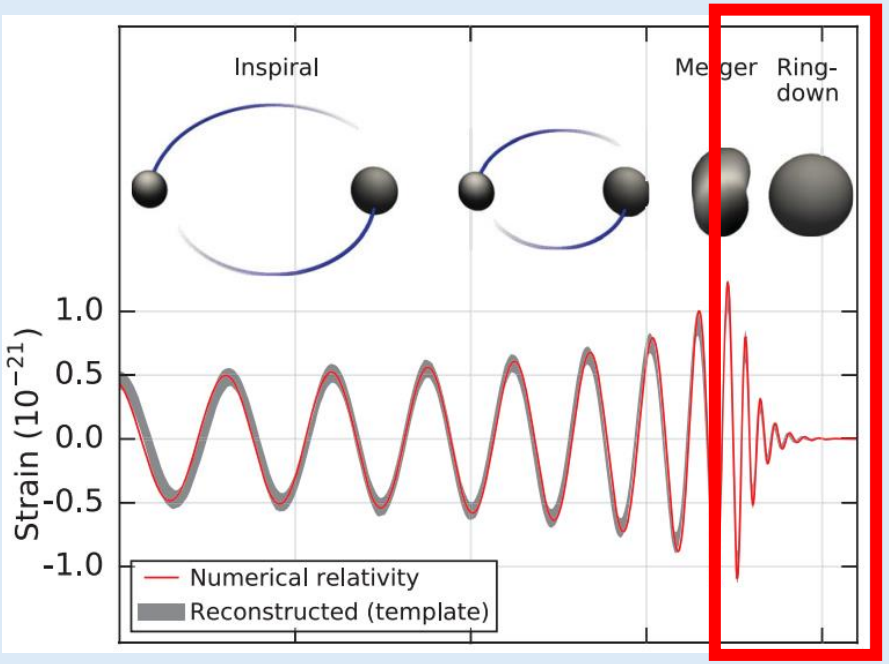
Gravitational Waveform Models

– Modeling the ringdown stage

- The ringdown GW can be modeled by a superposition of quasi-normal modes (QNMs) of the remnant BH.
  - ➢ QNM frequencies are uniquely determined by the remnant mass and spin. (derived within BH perturbation theory)



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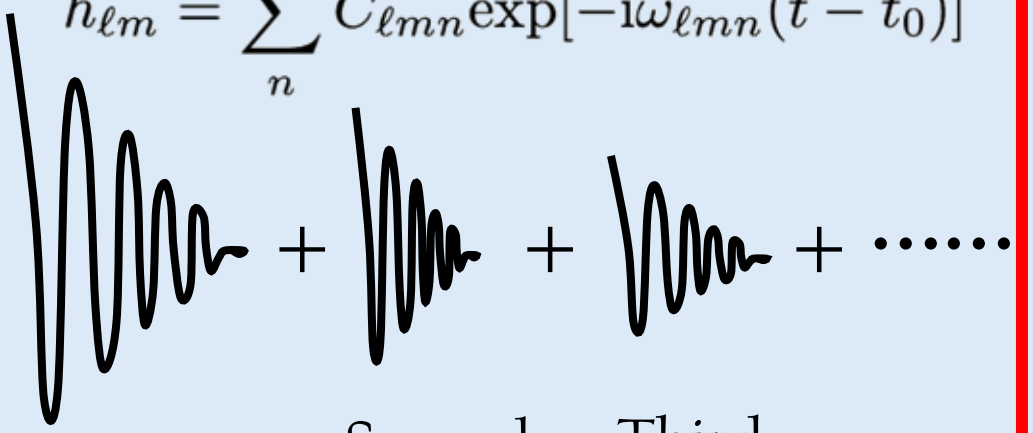
$$h_{\ell m} = \sum_n C_{\ell m n} \exp[-i\omega_{\ell m n}(t - t_0)]$$



First tone

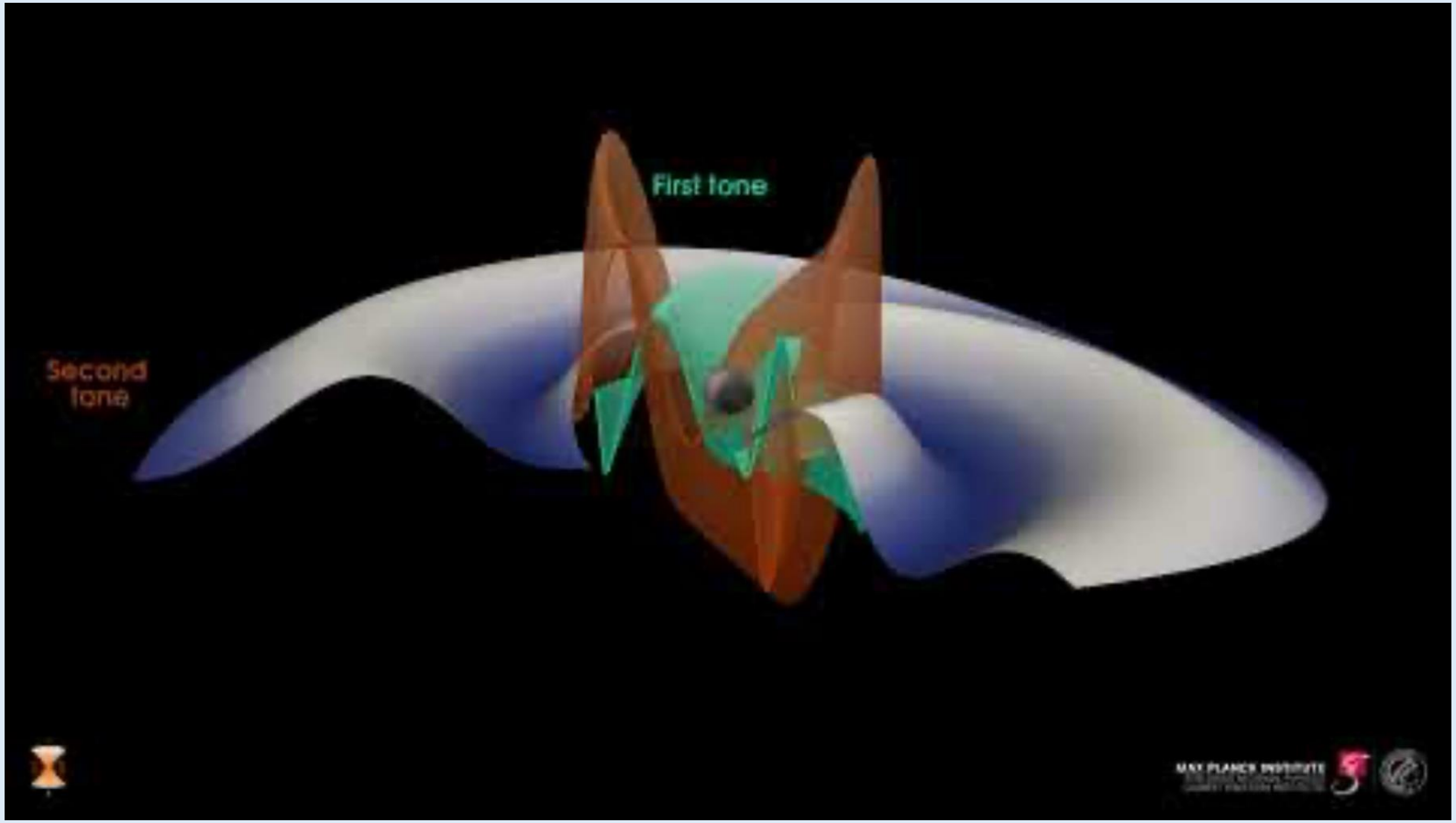
Second tone

Third tone

Gravitational Waveform Models

– Modeling the ringdown stage

- Visualization of the ringdown GW as a superposition of QNMs



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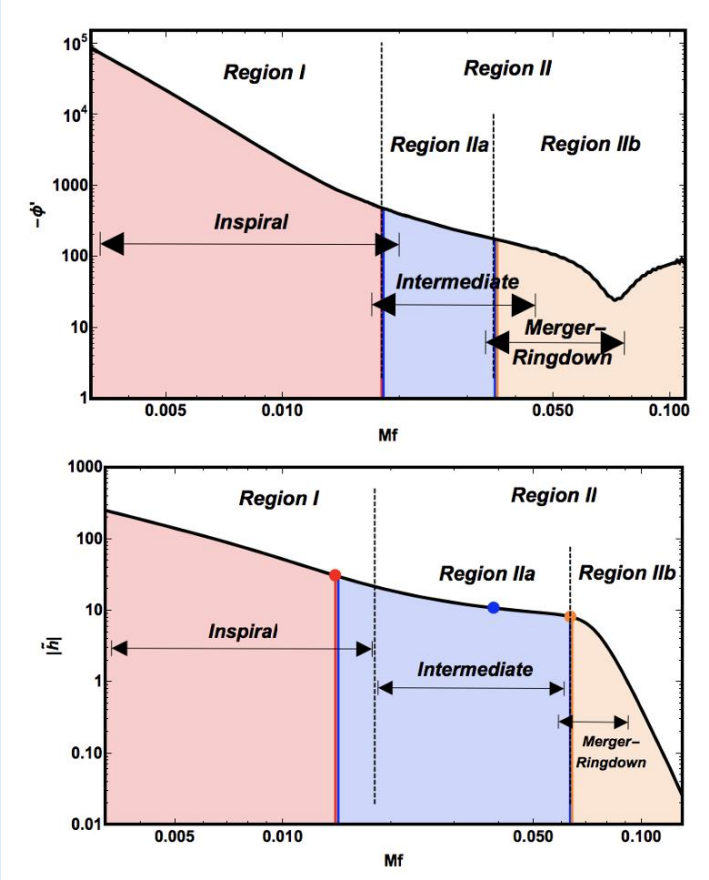
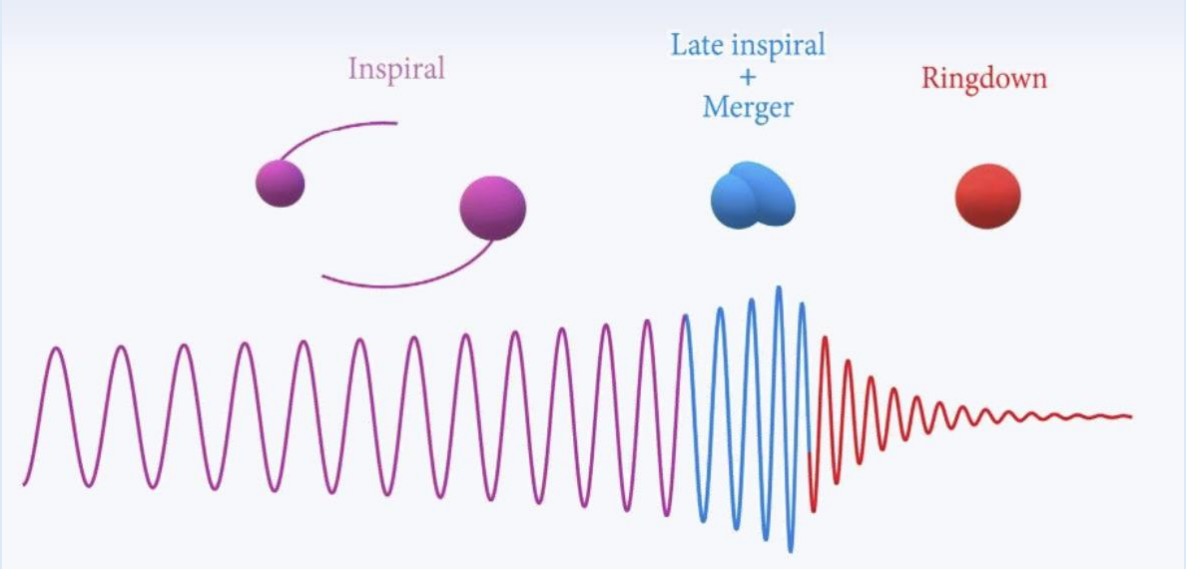
Gravitational Waveform Models

– Stitching inspiral, merger, and ringdown stages

- Create continuous inspiral–merger–ringdown (IMR) waveforms by stitching these 3 regimes.





Gravitational Waveform Models

– Phenomenological waveform models

- In GW data analysis, use phenomenological waveform models, which approximate the IMR waveform for continuous parameter space.
  - ➢ Numerical relativity simulation is computationally expensive for a single parameter configuration.

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– Phenomenological waveform models

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  - Numerical relativity simulation is computationally expensive for a single parameter configuration.

- Main waveform model families: IMRPhenom, EOB, and NRsurrogate.
  - IMRPhenom: frequency-domain
  - EOB (effective-one-body formalism): time-domain
  - NRsurrogate: time-domain, numerical relativity-based interpolation

Lecture 3: Parameter Estimation
– Content

- Gravitational waveform models
- Bayesian Inference

Bayesian Inference
– Bayes' theorem

- Bayesian inference is a framework that infers probability distributions of parameters from data.
  - ➢ Bayes' theorem

$$p(\theta|d,M) = \frac{\pi(\theta)L(d|\theta,M)}{Z}$$

$p(\mathrm{A}|\mathrm{B})$:
conditional probability of A given B (being true)

θ: parameters, d: observational data, M: GW model

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(or posterior for short)

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Prior probability distribution
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Likelihood function

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Evidence

$p(\mathrm{A}|\mathrm{B})$:
conditional probability of A given B (being true)

θ: parameters, d: observational data, M: GW model

Bayesian Inference
– Simple picture

- Update our knowledge about parameters using data within a model.
  - ➢ From prior to posterior based on Likelihood.

$$p(\theta|d,M) \propto \pi(\theta)L(d|\theta,M)$$

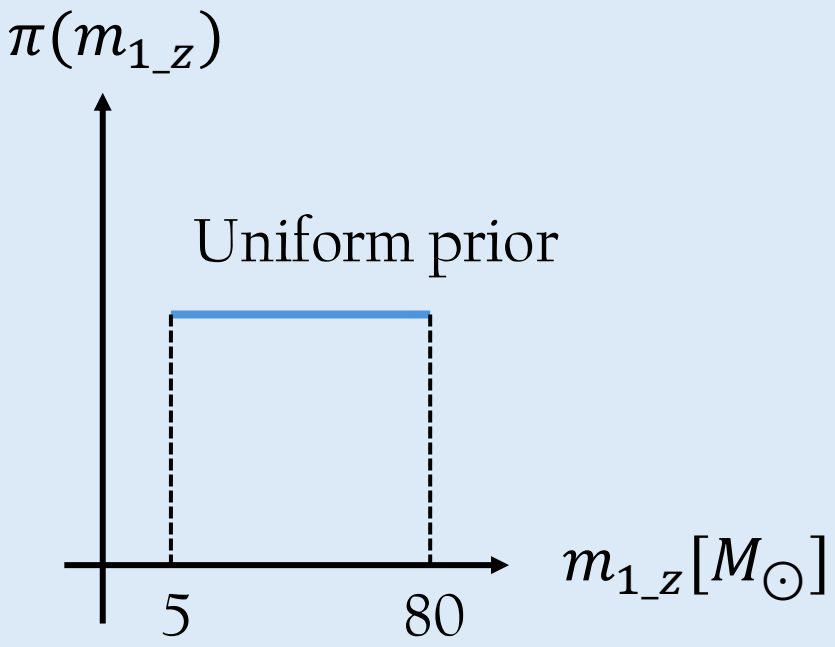
Posterior Prior likelihood

Update knowledge !!!

Bayesian Inference
– Prior probability distribution

- Prior $\pi(\theta)$ is our knowledge about parameters θ BEFORE observing the data.
  - ➢ Uniform priors are commonly used, but physical constraints are imposed, such as positive masses and bounded spins ($|\chi_{1,2}| < 1$).

Parameter	Range	Prior
$m_{1_z,2_z}$ $[M_\odot]$	[5, 80]	uniform
$q := m_2/m_1$	[0.125, 1]	set the bounds
$\mathcal{M}_{c_z}$ $[M_\odot]$	[20(5), 100]	set the bounds
$\chi_{1,2}$	[−0.99, 0.99]	uniform
d_L [Mpc]	[50, 2000]	uniform in volume
t_c [s]	[-0.1, 0.1]	uniform
ϕ_c	[0, 2π]	uniform
α	[0, 2π]	uniform
δ	[0, π]	uniform in $\cos\delta$
ι	[0, π]	uniform in $\sin\iota$
ψ	[0, 2π]	uniform



Bayesian Inference
– Likelihood

- The likelihood $L(d|\theta, M)$ measures how well a model M with given parameters θ explains the observed data.

➢ In GW data analysis, under assumption of stationary Gaussian noise, $L(d|\theta, M)$ takes the form,

$$L(d|\theta, M) \propto \exp\left[-\frac{(d - h(\theta)|d - h(\theta))}{2}\right]$$

$h(\theta)$: waveform predicted by model M

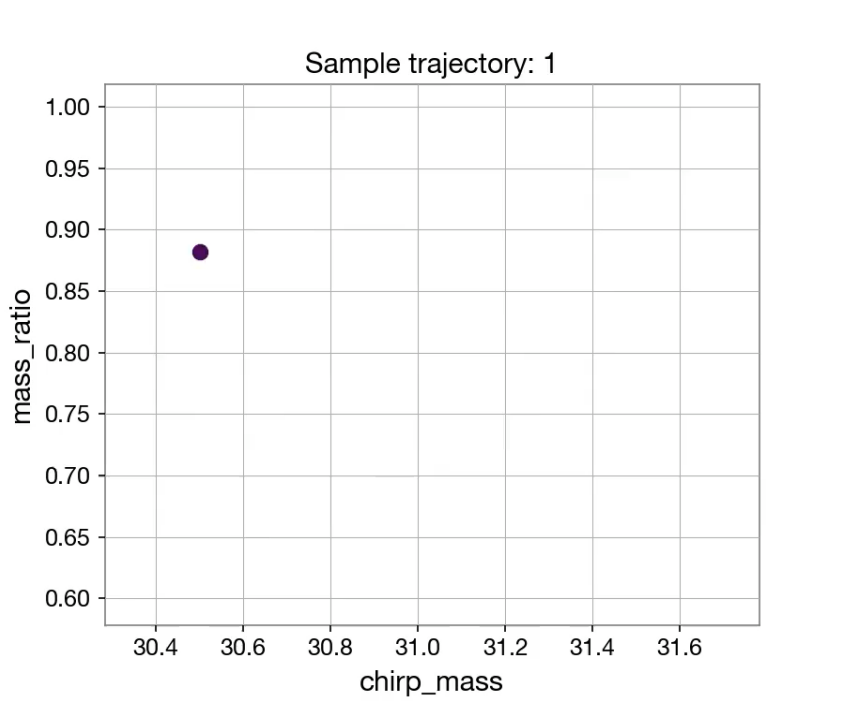
➢ $L(d|\theta, M)$ becomes large when the residual $d - h(\theta)$ is small.

Bayesian Inference
– Posterior

- Posterior $p(\theta|d, M)$ is our knowledge of parameter θ AFTER observing the data.

$$p(\theta|d, M) \propto \pi(\theta) L(d|\theta, M)$$

- ➢ Instead of computing the posterior directly, we sample it by using methods such as MCMC or nested sampling.
- ➢ The posterior is approximated by the distribution of samples.

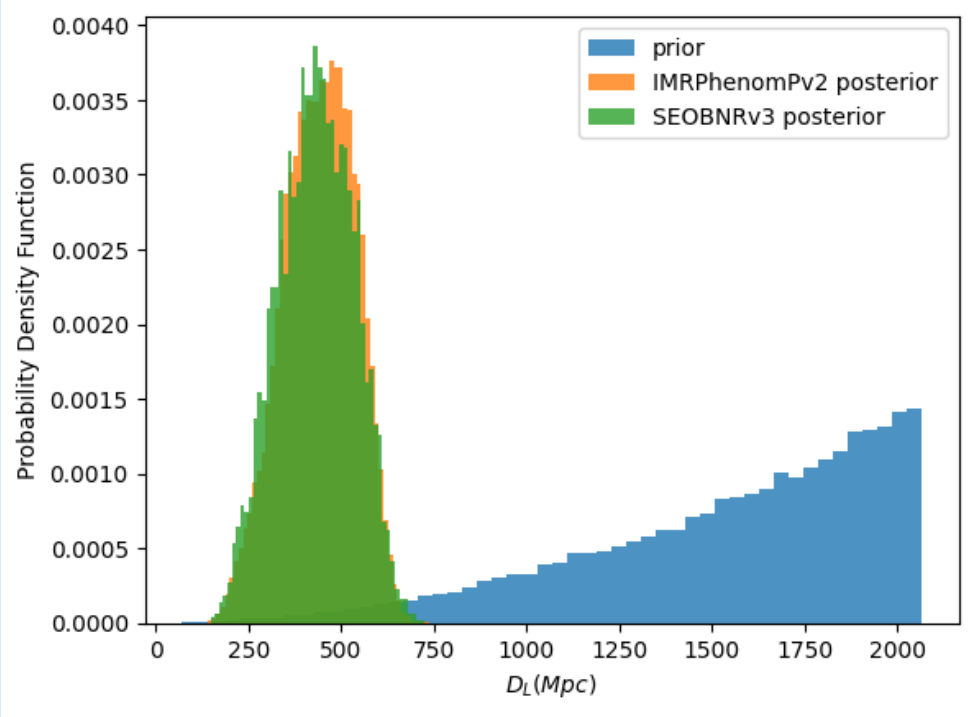


From tutorial 5.2 result

Bayesian Inference
– Posterior

- Tutorial 5.3: Check the results of GW150914

Posterior of luminosity distance of GW150914



From tutorial 5.3

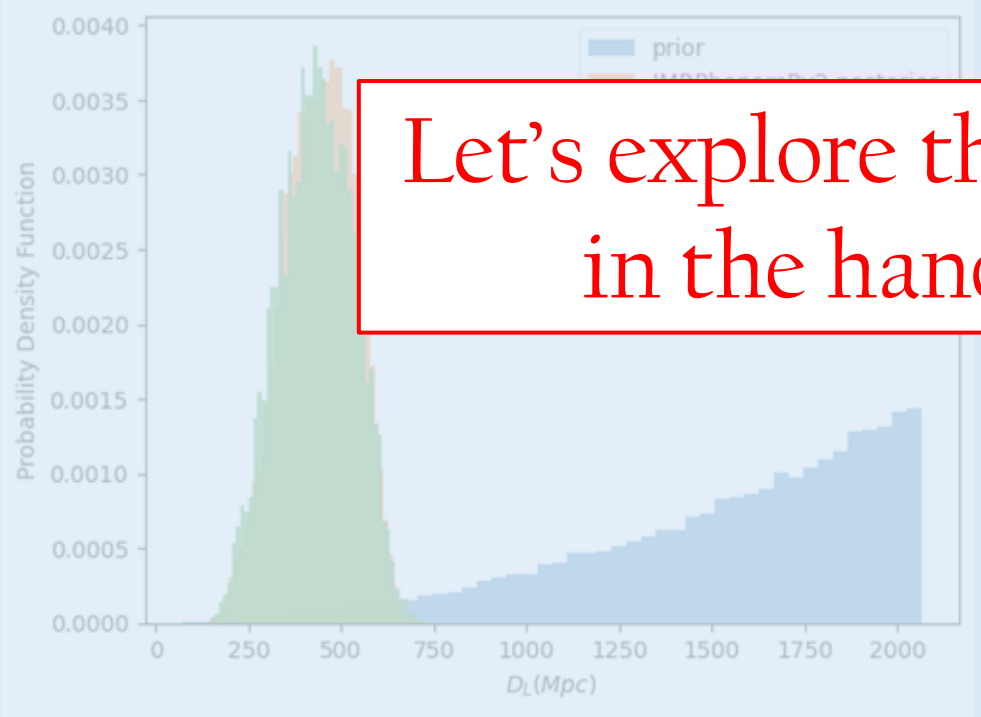
The data update our knowledge via likelihood, from the prior to the posterior !

The prior assumes a uniform distribution in comoving volume.

Bayesian Inference
– Posterior

- Tutorial 5.3: Check the results of GW150914

Posterior of luminosity distance of GW150914



From tutorial 5.3

Let's explore the other parameters in the hands-on session !

knowledge
e prior to
the posterior !

The prior assumes uniform distribution in comoving volume.

Summary of Lecture 3

- Accurate waveform models are essential for parameter estimation.
- Compact binary coalescences are commonly divided into three stages: the inspiral, merger, and ringdown.
- Bayesian inference updates our knowledge from the prior to the posterior via likelihood.
- The posterior is explored using sampling methods such as MCMC or nest sampling.